

WORLDS OF SCIENCE FICTION

SEPTEMBER 1952

35 cents

if



SHOCK TREATMENT

By Stanley Mullen

Also ROBERT MOORE WILLIAMS • CHARLES BEAUMONT
L. SPRAGUE DE CAMP • ALAN L. NOURSE • L. MAJOR WILLIAMS

WORLDS OF SCIENCE FICTION

SEPTEMBER 1952

35 cents

if



SHOCK TREATMENT

By Stanley Mullen

Also ROBERT MOORE WILLIAMS • CHARLES BEAUMONT
L. SPRAGUE DE CAMP • ALAN L. NOURSE • L. MAJOR WILLIAMS

The Project Gutenberg eBook of An Empty Bottle

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: An Empty Bottle

Author: Mari Wolf

Release date: March 11, 2010 [eBook #31601]

Language: English

Other information and formats: www.gutenberg.org/ebooks/31601

Credits: Produced by Sankar Viswanathan, Greg Weeks, and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK AN EMPTY
BOTTLE ***

Transcriber's Note:

This etext was produced from If Worlds of
Science Fiction September 1952. Extensive

research did not uncover any evidence that the U.S. copyright on this publication was renewed.



They wanted to go home—back to the planet they'd known. But even the stars had changed. Did the fate of all creation hinge upon an—

AN EMPTY BOTTLE

By Mari Wolf

Hugh McCann took the last of the photographic plates out of the developer and laid them on the table beside the others. Then he picked up the old star charts—Volume 1, Number 1—maps of space from various planetary systems within a hundred light years of Sol. He looked around the observation room at the others.

"We might as well start checking."

The men and women around the table nodded. None of them said anything. Even the muffled conversation from the corridor beyond the observation room ceased as the people stopped to listen.

McCann set the charts down and opened them at the first sheet—the composite map of the stars as seen from Earth. "Don't be too disappointed if we're wrong," he said.

Amos Carhill's fists clenched. He leaned across the table. "You still don't believe we're near Sol, do you? You're getting senile, Hugh! You know the mathematics of our position as well as anybody."

"I know the math," Hugh said quietly. "But remember, a lot of our basics have already proved themselves false this trip. We can't be sure of anything. Besides, I think I'd remember this planet we're on if we'd ever been here before. We visited every planetary system within a hundred light years of Sol the first year."

Carhill laughed. "What's there to remember about this hunk of rock? Tiny, airless, mountainless—the most monotonous piece of matter we've landed on in years."

Hugh shrugged and turned to the next chart. The others clustered around him, checking, comparing the chart with the photographic plates of their

position, finding nothing familiar in the star pattern.

"I still think we would have remembered this planet," Hugh said. "Just because it *is* so monotonous. After all, what have we been looking for, all these years? Life. Other worlds with living forms, other types of evolution, types adapted to different environments. This particular planet is less capable of supporting life than our own Moon."

Martha Carhill looked up from the charts. Her face was as tense and strained as her husband's, and the lines about her mouth deeply etched. "We've got to be near Earth. We've just got to. We've got to find people again." Her voice broke. "We've been looking for so long—"

Hugh McCann sighed. The worry that had been growing in him ever since they first left the rim of the galaxy and turned homeward deepened into a nagging fear. He didn't know why he was afraid. He too hoped that they were near Earth. He almost believed that they would soon be home. But the others, their reactions—He shook his head.

They no longer merely hoped. With them, especially with the older, ones, it was faith, a blind, unreasoning, fanatic faith that their journey was almost over and they would be on Earth again and pick up the lives they had left behind fifty-three years before.

"Look," Amos Carhill said. "Here are our reference points. Here's Andromeda Galaxy, and the dark nebula, and the arch of our own Milky Way." He pointed to the places he had named on the plates. "Now we can check some of these high magnitude reference stars with the charts."

Hugh let him take the charts and go through them, checking, rejecting. Carhill was probably right. He'd find Sol soon enough.

It had been too long for one shipful of people to follow a quest, especially a hopeless one. For fifty-three years they had scouted the galaxy, looking for other worlds with life forms. A check on diverging evolutions, they had called it—uncounted thousands of suns without planets, bypassed. Thousands of planetary systems, explored, or merely looked at and rejected. Heavy, cold worlds with methane atmospheres and lifeless rocks without atmospheres and even earth-sized, earth-type planets, with oceans and oxygen and warmth. But no life. No life anywhere.

That was one of the basics they had lost, years ago—their belief that life would arise on any planet capable of supporting it.

"We could take a spectrographic analysis of some of those high magnitude stars," Carhill said. Then abruptly he straightened, eyes alight, his hand on the last chart. "We don't need it after all. Look! There's Sirius, and here it is on the plates. That means Alpha Centauri must be—"

He paused. He frowned and ran his hand over the plate to where the first magnitude star was photographed. "It must be. Alpha Centauri. It has to be!"

"Except that it's over five degrees out of position." Hugh looked at the plate, and then at the chart, and then back at the plate again. And then he knew what it was that he had feared subconsciously all along.

"You're right, Amos," he said slowly. "There's Alpha Centauri—about twenty light years away. And there's Sirius, and Arcturus and Betelgeuse and all the others." He pointed them out, one by one, in their unfamiliar locations on the plates. "But they're all out of position, in reference to each other."

He stopped. The others stared back at him, not saying anything. Little by little the faith began to drain out of their eyes.

"What does it mean?" Martha Carhill's voice was only a whisper.

"It means that we discarded one basic too many," Hugh McCann said. "Relativity. The theory that our subjective time, here on the ship, would differ from objective time outside."

"No," Amos Carhill said slowly. "No, it's a mistake. That's all. We haven't gone into the future. We can't have. It isn't possible that more time has elapsed outside the ship than—"

"Why not?" Hugh said softly. "Why not millions of years? We've exceeded the speed of light, many times."

"Which disproves that space-time theory in itself!" Carhill shouted.

"Does it?" Hugh said. "Or does it just mean we never really understood space-time at all?" He didn't wait for them to answer. He pointed at the small, far from brilliant, star that lay beyond Alpha Centauri on the plates. "That's probably Sol. If it is, we can find out the truth soon enough."

He looked at their faces and wondered what their reactions would be, if the truth was what he feared.

The ship throbbed softly, pulsating in the typical vibrations of low speed drive. In the forward viewscreens the star grew larger. The people didn't look at it very often. They moved about the corridors of the ship, much as they usually moved, but quietly. They seemed to be trying to ignore the star.

"You can't be sure, Hugh." Nora McCann laid her hand on her husband's arm.

"No, of course I can't be sure."

The door from their quarters into the corridor was open. Several more people came in—young people who had been born on the ship. They were talking and laughing.

"Would it be so hard on the young ones, Hugh? They've never seen the Earth. They're used to finding nothing but lifeless worlds everywhere."

One of the young boys in the hall looked up at the corridor viewscreen and pointed at the star and then shrugged. The others turned away, not saying anything, and after a minute they left and the boy followed them.

"There's your answer," Hugh McCann said dully. "Earth's a symbol to them. It's home. It's the place where there are millions more like us. Sometimes I think it's the only thing that has kept us sane all these years—the knowledge that there is a world full of people, somewhere, that we're not alone."

Her hand found his and he gripped it, almost absently, and then he looked up at their own small viewscreen. The star was much bigger now. It was

already a definite circle of yellow light.

A yellow G-type sun, like a thousand others they had approached and orbited around and left behind them. A yellow sun that could have been anywhere in the galaxy.

"Hugh," she said after a moment, "do you really believe that thousands of years have gone by, outside?"

"I don't know what to believe. I only know what the plates show."

"That may not even be Sol, up ahead," she said doubtfully. "We may be in some other part of space altogether, and that's why the charts are different."

"Perhaps. But either way we're lost. Lost in space or in time or in both. What does it matter?"

"If we're just lost in space it's not so—so irrevocable. We could still find our way back to Earth, maybe."

He didn't answer. He looked up at the screen and the circle of light and his lips tightened. Whatever the truth was, they didn't have long to wait. They'd be within gravitational range in less than an hour.

He wondered why he was reacting so differently from the others. He was just as afraid as they were. He knew that. But he wasn't fighting the thought that perhaps they had really traveled out of their own time. He wondered what it was that made him different from the other old ones, the ones like Carhill who refused even to face the possibility, who insisted on clinging to their illusions in the face of the photographic evidence.

He didn't think that he was a pessimist. And yet, after only three years of their trip, after only fifty Earthlike but lifeless worlds, he had been the first to consider the possibility that life was unique to Earth and that their old theories concerning its spontaneous emergence from a favorable environment might be wrong.

Only Nora had agreed with him then. Only Nora could face this possibility with him now. The two of them were very much alike in their outlooks. They were both pragmatists.

But this time there would be no long years during which the others could slowly shift their opinions, slowly relinquish their old beliefs and turn to new ones. The yellow sun was too large and urgent in the screen.

"Hugh!"

He turned to the door and saw Amos Carhill standing there, bracing himself against the corridor wall. There was no color at all in Carhill's face.

"Come on up to the control room with me, Hugh. We're going to start decelerating any minute now."

Hugh frowned. He would prefer to stay and watch their approach on the screen, with Nora at his side. He had no duties in the control room. He was too old to have any part in the actual handling of the ship. Amos was old, too. But they would be there, all the old ones, looking through the high powered screens for the first clear glimpse of the third planet from the sun.

"All right, Amos." Hugh got up and started for the door.

"I'll wait here for you, Hugh," Nora said.

He smiled at her and then followed Carhill out into the crowded corridor. No one spoke to them. Most of the people they passed were neither talking, nor paying any attention to anything except the corridor screens, which they could no longer ignore. The few who were talking spoke about Earth and how wonderful it would be to get home again.

"You're wrong, Hugh," Amos said suddenly.

"I hope I am."

The crowd thinned out as they passed into the forward bulkheads. The only men they saw now were the few young ones on duty. Except for their set, anxious faces they might have been handling any routine landing in any routine system.

The ship quivered for just a second as it shifted over into deceleration. There was an instant of vertigo and then it was gone and the ship's gravity felt as normal as ever. Hugh didn't even break stride at the shift.

He followed Carhill to the control room doorway and pushed his way in, taking a place among the others who already clustered about the great forward screen. The pilot ignored them and worked his controls. The screen cleared as the ship's deceleration increased. The pilot didn't look at it. He was a young man. He had never seen the Earth.

"Look!" Amos Carhill cried triumphantly.

The screen focused. The selector swung away from the yellow sun and swept its orbits. The dots that were planets came into focus and out again. Hugh McCann didn't even need to count them, nor to calculate their distance from the sun. He knew the system too well to have any trouble recognizing it.

The sun was Sol. The third planet was the double dot of Earth and moon. He realized suddenly that he had more than half expected to see an empty orbit.

"It's the Earth all right," Carhill said. "We're home!"

They were all staring at the double dot, where the selector focused sharply now. Hugh McCann alone looked past it, at the background of stars that were strewn in totally unfamiliar patterns across the sky. He sighed.

"Look beyond the system," he said.

They looked. For a long time they stared, none of them speaking, and then they turned to Hugh, many of them accusingly, as if he himself had rearranged the stars.

"How long have we been gone?" Carhill's voice broke.

Hugh shook his head. The star patterns were too unfamiliar for even a guess. There was no way of knowing, yet, how long their fifty-three years had really been.

Carhill shook his head, slowly. He turned back to the screen and stared at the still featureless dot that was the Earth. "We can't be the only ones left," he said.

No one answered him. They were still stunned. They couldn't even accept, yet, the strange constellations on the screen.

End of the voyage. Fifty-three years of searching for worlds with life. And now Earth, under an unfamiliar sky, and quite possibly no life at all, anywhere, except on the ship.

"We might as well land," McCann said.

The ship curved away from the night side of the Earth and crossed again into the day. They were near enough so that the planetary features stood out sharply now, even through the dense clouds that rose off the oceans. But although the continental land masses and the islands were clearly defined, they were as unrecognizable as the star constellations had been.

"That must be North America," Amos Carhill said dully. "It's smaller than the continent on the night side...."

"It might be anywhere," Hugh McCann said. "We can't tell. The oceans look bigger too. There's less land surface."

He stared down at the topography thousands of miles below them. Mountains rose jaggedly. There were great plains, and crevasses, and a rocky, lifeless look everywhere. No soil. No erosion, except from the wind and the rains.

"There's no chlorophyll in the spectrum," Haines said. "It seems to rule out even plant life."

"I don't understand." Martha Carhill turned away from the screen. "Everything's so different. But the moon looked just exactly like it always did."

"That's because it has no atmosphere," Hugh said. "So there's no erosion. And no oceans to sweep in over the land. But I imagine that if we explored it we'd find changes. New craters. Maybe even new mountains by now."

"How long has it been?" Carhill whispered. "And even if it's been millions of years, what happened? Why aren't there any plants? Won't we find anything?"

"Maybe there was an atomic war," the pilot said.

"Maybe." Carhill had thought of that too. Probably all of them had. "Or maybe the sun novaed."

No one answered him. The concept of a nova and then of its dying down, until now the sun was just as it had been when they left, was too much.

"The sun looks hotter," Carhill added.

The ship dropped lower, its preliminary circle of the planet completed. It settled in for a landing, just as it had done thousands of times before. And the world below could have been any of a thousand others.

They dropped quickly, braking through the atmosphere, riding it down. The topography came up to meet them and the general features blurred, leaving details standing out sharply, increasing in sharpness as if the valleys and mountains below were tiny microscopic crystals under a rapidly increasing magnification.

The pilot picked their landing place without difficulty. It was a typical choice, a spot on the broad shelving plain at the edge of the ocean. The type of base from which all tests on a planet could be run quickly, and a report written up, and the files of another world closed and tagged with a number and entered in one of the great storage encyclopedias.

Even to Hugh there was an air of unreality about the landing, as if this planet wasn't really Earth at all, despite its orbit around the sun, despite its familiar moon. It looked too much like too many others.

The actual landing was over quickly. The ship quivered, jarred slightly, and then was still, resting on the gravelled plain that had obviously once been part of the ocean bed. The ocean itself lay only a few hundred yards away.

Hugh McCann looked out through the viewscreen, turned to direct vision now. He stared at the waves swelling against the shore and his sense of

unreality deepened. Even though this was what he had more than half expected, he couldn't quite accept it, yet.

"We might as well go out and look around," he said.

"Air pressure, Earth-norm." Haines began checking off the control panel by rote. "Composition: oxygen, nitrogen, water vapor—"

"There's certainly nothing out there that could hurt us," Martha Carhill snapped. "What could there be?"

"We might check for radioactivity," Hugh said quietly.

She turned and stared at him. Her mouth opened and then snapped shut again.

"No," Haines said. "There's no radioactivity either. Everything's clear. We won't need space suits."

He pressed the button that opened the inner locks.

Carhill glanced over at him and then switched on the communicator, and the noises from the rest of the ship flooded into the control room. Everywhere people were milling about. Snatches of talk drifted in, caught up in the background as various duty officers, reported clearance on the landing. Most of the background voices were young, talking too loudly and with too much forced cheerfulness about what lay outside the ship.

Hugh sighed, as aware of all the people as if he were out in the corridors with them. It was the space-born ones who were doing most of the talking. The children, the young people, the people no longer young but still born since the voyage started, still looking upon Earth more as a wonderful legend than as their own place of origin.

The old ones, those who had left the Earth in their own youth, had the least of all to say. They knew what was missing outside. The younger ones couldn't really know. Even the best of the books and the pictures and the

three dimensional movies can give only a superficial idea of what a living world is like.

"Hugh." Carhill clutched his arm.

"Yes, Amos."

"There must be people, somewhere. There have to be. Our race can't be dead."

Hugh McCann looked past him, out at the sky and the clouds of water vapor that swirled up to obscure the sun. The stars, of course, were completely hidden in the daylight.

"If there are any others, Amos, we can be pretty certain they're not on Earth."

"They may have left. They may have gone somewhere else."

"No!" Martha Carhill's face twisted and then went rigid. "There's no one anywhere. There can't be. It's been too long. You saw the stars, Amos—the stars—all wrong, every one of them!"

Her hands came up to her face and she started to cry. Amos crossed over to her and put his arms around her.

Hugh McCann watched them for a moment and then he turned and left them and went out through the locks after the young people. He didn't know what to think. He wished that they had never turned back to Earth at all, that they had kept going, circling around the rim of the galaxy forever.

He went through the outer lock and then down the ramp to the ground.

He stood on the Earth again, for the first time since his early youth. And it was not the same. There was bare rock under his feet and bare rock all around him, gravel and boulders and even fine grained sand. But no dust. No dirt. No trace of anything organic or even ever touched by anything organic.

He had walked too many worlds like this. Too many bare gray worlds with bare gray oceans and clouds of vapor swirling up into the warm air. Too

many worlds where there was wind and sound and surf; where there should have been life, but wasn't.

This was just another of those worlds. This wasn't Earth. This was just a lifeless memory of the Earth he had known and loved. For fifty-three years they had clung to the thought of home, of people waiting for them, welcoming them back someday. Fifty-three years, and for how many of those ship-years had Earth lain lifeless like this?

He looked up at the sky and at all the stars that he couldn't see and he cursed them all and cursed time itself and then, bitterly, his own fatuous stupidity.

The people came out of the ship and walked about on the graveled plain, alone or in small groups. They had stopped talking. They seemed too numbed by what they had found to even think, for a while.

Shock, Hugh McCann thought grimly. First hysteria and tears and loud unbelief, and now shock. Anything could come next.

He stood with the warm wind blowing in his face and watched the people. In the bitter mood that gripped him he was amused by their reactions. Some of them walked around aimlessly, but most, those who were active in the various departments, soon started about the routine business of running tests on planetary conditions. They seemed to work without thinking, by force of habit, their faces dazed and uncaring.

Conditioning, Hugh thought. Starting their reports. The reports that they know perfectly well no one will ever read.

He wandered over to where several of the young men were sending up an atmosphere balloon and jotting down the atmospheric constituents as recorded by the instruments.

"How's it going?" he said.

"Earth-norm. Naturally—" The young man flushed.

"Temperature's up though. Ninety-three. And a seventy-seven percent humidity."

He left them and walked down across the rocks to the ocean's edge. Two young girls were down there before him, sampling the water, running both chemical and biological probing tests.

"Hello, Mr. McCann," the taller girl said dully. "Want our report?"

"Found anything?" He knew already that there was nothing to find. If there were life the instruments would have recorded its presence.

"No. Water temperature eighty-six. Sodium chloride four-fifths Earth normal." She looked up, surprised. "Why so low?"

"More water in the ocean, maybe. Or maybe we've had a nova since we were here last."

It was getting late, almost sunset. Soon it would be time for the photographic star-charts to be made. Hugh brought himself up short and smiled bitterly. He too was in the grip of habit. Still, why not? Perhaps they could estimate, somehow, how many millions of years had passed.

Why? What good would it do them to find out?

After a while the sun set and a little later the full moon rose, hazy and indistinct behind the clouds of water vapor. Hugh stared at it, watched it rise higher until it cleared the horizon, a great bloated bulk. Then he sighed and shook his head to clear it and started to work. The clouds were thick. He had to move the screening adjustment almost to its last notch before the vapor patterns blocked out and the stars were bright and unwavering and ready to be photographed. He inserted the first plate and snapped the picture of the stars whose names he knew but whose patterns were wrong, some subtly, some blatantly.

There was something he was overlooking. Some other factor, not taken into account. He developed the first plates and compared them with the star charts of Earth as it had been before they left it, and he shook his head. Whatever the factor was, it eluded him. He went back to work.

"Oh, here you are, Hugh."

He jumped at the sound of Carhill's voice. He had been working almost completely by habit, slowly swinging the telescope across the sky and snapping the plates. And trying to think.

"Why waste time on that?" Carhill added bitterly. "Who's ever going to see our records now?"

Behind Carhill, several of the other old ones nodded. Hugh was surprised that they had managed to come back to the ship without his hearing them. But of course they had come back in at sundown, as usual on a routine check, and now they were gathering to compile their reports. Hugh looked from face to face, wondering if he too was as numb and dazed and haggard appearing as they were. He probably was.

"What do you suggest, Amos?" he said.

"I say there's no use going on," Carhill said flatly. "You've all run your tests. And what have you found? No fossils. Not even a single-celled life form in the ocean. No way even to tell how many millions of years it's been."

"Maybe it hasn't been so long," Haines said. "Maybe something happened here fairly recently, and the people all went to some other system—to one of the Centauri planets, maybe."

Amos Carhill laughed bitterly. "You can say that in the face of the evidence? We *know* that millions of years have passed. Nothing's the same. Even the tides are three times what they were. It's obvious what happened. The sun novaed. Novaed and cooled. Do you really believe that our race has lasted that long, on some nearby system?"

His voice rose. He glared about at the others. He threw back his head suddenly and laughed, and the laughter echoed and re-echoed off the steel walls.

"I say let's die now!" Carhill cried. "There's no use going on. Hugh was right, as usual. We shouldn't have tried to come back. We've been fools, all these years, thinking we had a world to come home to."

The people muttered, crowded closer. They pushed into the observation room, shoved nearer to it in the outside corridor. They muttered in a rising note of panic as the numbing shock that gripped them gave way.

"Why not die here?" Martha Carhill's voice rose shrill above the sound of her husband's laughter. "We should have died here millions of years ago!"

Hugh McCann looked at her and at Amos and at all the others. He sighed. Why not? Why go on? There was no answer. Even a pragmatist gave up eventually, when the facts were all against him.

He glanced down at the reports on the table. All the routine reports, gathered together into routine form, written up in routine terminology. Reports on an Earth-type planet that just happened to be the Earth itself.

And then, quite suddenly, the obvious, satisfactory answer came to him. The factors clicked into place, and he wondered why he hadn't thought of them long ago. He looked up from the reports, at the people on the verge of panic, and he knew what to say to quiet them. He had the factors now.

"No!" he cried. "You're wrong. There's no reason at all to assume that our race is dead!"

Amos Carhill stopped laughing and stared at him and the others stared also and none of them believed him at all.

"It's simple!" he cried. "Why has so much time passed outside the ship while to us only fifty-three years have gone by?"

"Because we traveled too fast," Carhill said flatly. "That's why."

"Yes," Hugh said softly. "But there's one thing we've been forgetting. What we did, others could do also. Probably lots of expeditions started out after we left, all trying for the speed of light."

They stared at him. Slowly the dazed look died out of their eyes as they realized what he meant, and what the concept might mean to them. The concept of other ships, following them out into time. The concept of other men, also millions of years from the Earth they had left.

"You mean," Carhill said slowly, "that you believe other people got caught in the same trap we did—that there may be others *in this time also*?"

Hugh nodded. "Why not? Maybe they colonized some of those Earth-type planets we checked on. Anyway, we can look for them."

"No." Carhill shook his head. "If any of them had started after us we would have crossed their paths already. We never have. We never found a trace of any other expedition. Even if there is another, even if there are colonies somewhere, we could spend another fifty years looking."

"Well," Martha Carhill whispered. "Why not? It would give us something to look for."

Hugh McCann glanced around the circle of faces and saw the new hope that came into them, the new belief that sprang into existence so quickly because they wanted to believe. He smiled, somewhat sadly, and picked up the pile of reports and the photographs he had just developed. Then he slipped out of the room, through the crowd outside, away from them and the rising hum of their voices. He didn't need to say anything more. The ship would go on.

“**H**ugh, is that you?"

"Yes, Nora."

She was waiting for him in the corridor. She came up to him and smiled and slipped her arm through his. They walked on together, down the hall past the last of the people.

"I heard what you said, Hugh. You convinced them."

He nodded. "I wonder why it took me so long to think of it."

The voices died away behind them. They were all alone. They rounded a corner where a viewscreen picked up the image of the moon, so familiar, now the only thing that was familiar about this Earth. Nora shivered.

"You were very logical, Hugh. But I didn't believe you."

He glanced around and saw that there was no one near them and that the communicators in this part of the ship were turned off. Only then did he answer her.

"I didn't believe myself, Nora."

"Tell me."

"When we're outside."

They went down the winding ramp that led to the interior of the ship. It too was deserted now. They left the carpeted, muffled corridors and their footsteps rang on the steel plates that lay down the middle of the ship, its heart, where the energy converters were, and the disposal units, and the plant rooms, and the great glass spheres of the hydroponics tanks.

"It's ironic, isn't it?" Nora said slowly. "We left here so long ago, looking for worlds with life, and we come back to find our own world dead."

"It's ironic, all right." He walked along the row of tanks until he came to the one he was searching for, and then he picked up a glass cylinder and filled it from the tank.

"I had to tell them something, Nora. They couldn't have gone on, otherwise."

The bottle was full. He stoppered it and then turned away. They crossed to the nearest lock and he pushed the button that opened it. They waited a few minutes until the door came open, and then they went out, down the ramp to the ground, across the slippery rocks. Even through the clouds there was enough light to see by.

"It's warm," she said.

"It always is, now."

They were approaching the ocean. The surf beat loudly in their ears. The spray was warm against their faces, almost as warm as the night wind.

"Tell me," she said. "You know what really happened, don't you?"

"I think so. I can't really be sure."

They paused on the low ledge where he had stood earlier and watched the girls gather their data for the reports. At their feet the waves washed up to the edges of the tide pools, eddying into and out of them softly. The water looked dark and cold, but they knew that it too was warm.

"There've been lots of changes, and they all fit a pattern," he said. "The temperature. The difference in salt content in the water. The higher tides. Those things could happen for several reasons. But there's only one explanation for the other changes, the ones I found on the star charts."

She waited. The water lapped in and out, reaching almost to where they stood.

"The Earth rotates faster now," he said. "And the stars are nearer. Much nearer than they were."

"Isn't that impossible?"

"How do we know? We exceeded the speed of light. Who could say what continuum that might have put us in? I remember an analogy I read once, in a layman's book on different theories of space-time. '—The future and the past, two branches of a hyperbola, each with the speed of light as its limit —'"

"You mean," she whispered, "that we're not in the future at all? We're in the past—the far past—before there was any life on Earth?"

He looked down at the pools of water at their feet, the lifeless water that according to all their old discarded theories should have been teeming with life. He nodded slowly and lifted the glass cylinder he had brought from the ship and stared at it.

"That bottle," she whispered. "You filled it with bacteria, didn't you?"

He nodded again.

"You're mad, Hugh. You can't mean that *that bottle* is the origin of life on Earth! You can't."

"Maybe this isn't our Earth, Nora. Maybe there are thousands of continuums and thousands of Earths, all waiting for a ship to land someday and give them life."

Slowly he unstoppered the cylinder and knelt down at the water's edge. For a minute he paused, wondering if there were other continuums or only this one, wondering just how deep the paradox lay. Then he tipped the bottle up and poured, and the liquid from the cylinder ran down into the tide pools and eddied there and was lost in the liquid of the ocean. He poured until the bottle was empty and all the single-celled bacteria from the ship's tank mingled with the warm, lifeless waters.

The water temperatures were the same. Everything was the same, and the conditions were very favorable and the bacteria would divide and redivide and keep on dividing for millions of years.

"We'll hold the ship under light speed," he said. "And in a few million years we can drop back here and see how evolution is getting along."

He stood up and she took his hand and moved closer to him. They were both shivering, despite the warmth of the air.

"But how did life originate in the beginning?" she asked suddenly.

Hugh McCann shook his head in the darkness. "I don't know. We've been all over the galaxy and haven't found life anywhere. Perhaps it can't have a natural cause. Perhaps it's always planted. A closed circle from beginning to end."

"But something—someone—must have started the circle. Who?"

He looked down at the empty cylinder that he had dropped at the water's edge and then he looked out at the ocean, lifeless no longer. And once again he shook his head.

"We did, Nora. We're the beginning."

For a long moment their eyes met and held, and then they turned and walked away from the ocean, back toward the ship, and the people. And the moonlight glinted off the empty bottle.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK AN EMPTY
BOTTLE ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.